

Stress-Response Genes and the Hallmarks of Aging in Arthropods

Executive summary of a structured literature review | Capstone research project | Chi Leong Andy Fu

41 papers coded	22 species across crustaceans, insects and arachnids	51% of papers study HSP70	29% directly measure lifespan or age-related change	2 papers causally link a gene to longevity
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Bottom line

Arthropods offer a rich literature on how stress-response genes, above all heat shock proteins, help animals survive heat, infection, toxins and poor diets. But most of that literature links to aging only by inference. Fewer than a third of the 41 papers measure aging directly, and only two manipulate a gene and then show a change in longevity. The field is well placed to test whether declining stress-response capacity drives aging in invertebrates, yet that test has mostly not been run.

Scope and approach

Each paper was coded into a structured database recording species, stress type, genes studied, gene function, experimental method, key findings, and the hallmark(s) of aging it bears on, using the López-Otín hallmarks framework. The database was then cleaned: category labels were standardised, a duplicated species synonym was merged, and multi-valued fields were split so they could be counted. Three derived fields were added to support synthesis. **Gene family** groups the genes (for example HSP70, HSP90, small HSPs, HSF1, antioxidant enzymes). **Evidence type** grades each paper as causal (knockdown, silencing, recombinant protein), correlational (expression change only) or physiological biomarker. **Direct aging measure** records whether the paper itself measures lifespan or age-related change, rather than the link being argued by the reviewer.

Key findings

- 1. The evidence base is concentrated on one pathway.** HSP70 appears in 21 of 41 papers, and loss of proteostasis is the hallmark in 29 (71%). Mitochondrial dysfunction/oxidative stress (11) and altered intercellular communication, mostly immune function (12), follow. Genomic instability, epigenetic change and nutrient sensing have two or three papers each; telomere attrition and cellular senescence are absent.
- 2. The link to aging is mostly inferred, not measured.** 58% of papers are correlational, typically reporting that a stress gene is upregulated under a challenge. Only 12 papers measure aging directly. The two that combine a genetic manipulation with a longevity readout are ArHsp40 knockdown in brine shrimp (*Artemia franciscana*), which shortened lifespan, and loss of HSF-1 in the copepod *Tigriopus californicus*, which accelerated aging.
- 3. Two literatures that rarely meet.** An older physiological literature in flies measures aging directly (lifespan, lipid peroxidation, age pigments, rate-of-living) but studies few genes. A newer molecular literature, dominated by *Artemia* and farmed crustaceans (12 papers from aquaculture species), studies genes in depth but seldom follows animals across their lifespan.
- 4. HSF1 stands out as a mechanistic lead.** In *Daphnia*, the short-lived species loses its ability to induce Hsp70 by mid-life while the long-lived species keeps it, and this tracks a decline in HSF-1 DNA binding. Together with the two causal HSF1 studies, this points to reduced stress-response inducibility, not simply lower gene expression, as the variable most worth testing.
- 5. Hormesis recurs across very different systems.** Mild non-lethal heat shock protects shrimp against *Vibrio* and white spot virus through HSP70/HSP90, and low-dose radiation extends fruit fly lifespan through HSP and DNA-repair pathways. Dietary supplements in aquaculture that raise HSP70 and SOD fit the same pattern of mild stress priming resilience.

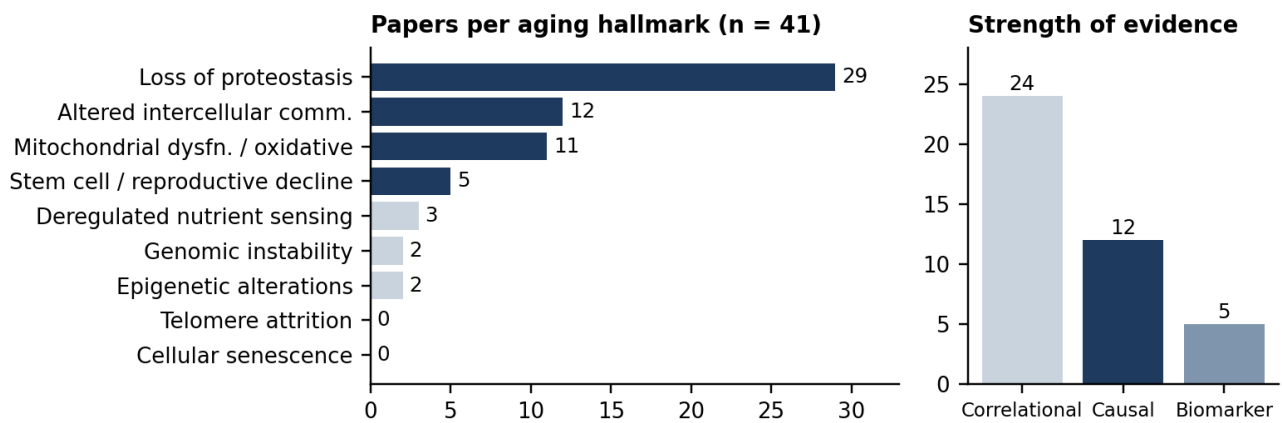


Figure 1. Left: coverage of aging hallmarks (a paper can map to several; telomere attrition and cellular senescence shown for reference). Right: papers by strength of evidence.

Gaps and recommended next steps

Close the causal gap. The strongest next study would combine gene knockdown (HSP70, HSP40 or HSF1) with a full lifespan assay in a tractable species such as *Artemia*, where RNAi already works and diapause offers a natural model of arrested aging.

Measure inducibility over age. Repeating the *Daphnia* design, testing stress-response induction in young versus old animals, in other taxa would show whether loss of inducibility is general.

Broaden the hallmarks. Genomic, epigenetic and nutrient-sensing mechanisms are barely covered; the one transgenerational epigenetic study (phloroglucinol in *Artemia*) suggests this is a productive direction.

Use aquaculture data. Priming and diet trials in farmed species already generate stress-gene data; adding survival-over-time endpoints would make them informative for aging research.

Limitations

The review covers 41 papers and is weighted toward crustaceans (54%). Hallmark assignments, evidence grades and the direct-aging flag are reviewer judgements, and three papers carry non-standard hallmark labels still to be resolved. Counts describe the coded literature, not effect sizes, and publication year was not recorded, so trends over time cannot yet be assessed.